

LINTANG SYAWALANI HAWA NINGRUM

085290151439 | syawalanilintang@gmail.com | linkedin.com/in/lintangsyawalani | https://github.com/lintangsyawalani/Portofolio-Lintang
Surabaya, Indonesia

Fresh graduate in Mathematics from Institut Teknologi Sepuluh Nopember (ITS) with a strong interest in data analytics, machine learning, and statistical analysis. Experienced in data collection, preprocessing, validation, analysis, forecasting, and reporting through academic projects, organizational activities, and an internship at Statistics Indonesia (BPS) Surakarta. Developed a final-year project on phishing email classification using stylometric features, Support Vector Machine (SVM), and Explainable AI (SHAP), involving the processing and analysis of tens of thousands of email records. Also completed a forecasting project on Surakarta City's 2025 inflation rate using the ARIMA method based on Consumer Price Index (CPI) data during the internship. Proficient in Python, Microsoft Excel, SQL and Google Sheets for data processing and statistical analysis. Detail-oriented, analytical, and adaptable, with strong communication, teamwork, and organizational skills developed through leadership roles in student organizations.

TECHNICAL SKILLS

Programming Language : Python, SQL Server, Java

Software and Productivity : Microsoft Office (Excel, Word, & Power Point), Google Sheets, LaTeX, Github

Libraries & Tools : NumPy, Pandas, Scikit-learn, NLTK, Git, Matplotlib, Seaborn

Languages : Bahasa Indonesia (Fluent), English (Intermediate – TOEFL Score 477)

PROJECTS

Phishing Email Classification System Based on Stylometric Features Using Support Vector Machine and Shapley Additive Explanations

- Extracted 60 text stylometric features (including lexical, syntactic, punctuation characteristics, and readability scores) from email content to differentiate patterns between phishing and legitimate emails.
- Developed a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel optimized via GridSearchCV, achieving an accuracy of 88.87% and an F1-score 89.49%.
- Applied Shapley Additive Explanations (SHAP) for transparent interpretation of feature contributions, and implemented the model into an interactive Streamlit web application integrated with the Gmail API.

Surakarta Poverty Rate Forecasting (2025) using Time Series Linear Regression

- Developed time-series linear regression models using Python (OLS estimation) by engineering lag features (t-1, t-2), 2-year rolling means, time trends, and cross-lag indicators to forecast poverty metrics.
- Evaluated models on 2024 holdout data, achieving exceptional predictive accuracy with a MAPE of 0.28% for the poverty rate and 0.59% for the total poor population, outperforming baseline Naive and Mean models.
- Generated one-step-ahead forecasts for 2025 (projecting 8.243% poverty rate and 43.097k poor population), providing data-driven insights to support local socio-economic planning at BPS Surakarta.

GoClean: Smart Integrated Waste Pickup & Spatial Database System

- Designed and implemented a structured relational SQL Server database architecture covering user roles, pickup requests, pricing, dynamic transactions, and notifications.
- Integrated Geographic Information System (GIS) features enabling radius-based spatial search (using the Haversine formula and buffer visualization) alongside traditional keyword and district-based filtering.
- Modeled end-to-end digital waste collection workflows to streamline waste pickup logistics and enhance Temporary Waste Disposal Site (TPS) operational efficiency.

WORK EXPERIENCE

Statistics Indonesia (BPS) Surakarta - Surakarta, Indonesia

Jun 2025 - Aug 2025

Neraca Intern

- Collected, processed, and validated statistical data to ensure data accuracy and consistency.
- Compiled and managed datasets for economic indicator publications and statistical reports.
- Prepared infographics, presentation materials, and statistical documentation to support reporting activities.

- Assisted in processing regional budget realization (APBD) reports and maintaining administrative statistical records. Conducted a forecasting project on Surakarta City's 2025 inflation rate using the ARIMA method based on Consumer Price Index (CPI) data and presented the results to supervisors.

EDUCATION

Institut Teknologi Sepuluh Nopember - Surabaya, Indonesia

Jul 2022 - Aug 2026

Bachelor of Mathematics, 3.34/4.00

- Thesis: "Phishing Email Classification Based on Stylometric Features Using Support Vector Machine and Shapley Additive Explanations)."
- Relevant Coursework : Data Mining, Machine Learning, Artificial Intelligence, Pattern Recognition, Statistical Computing, Database Systems, Linear Algebra, Probability and Statistics, Natural Language Processing (Text Mining), Data Visualization.

SMA Negeri 1 Baturetno - Wonogiri, Indonesia

Jul 2019 - May 2022

High School IPA

ORGANIZATIONAL EXPERIENCE

Ikatan Mahasiswa Berprestasi Wonogiri

Jan 2025 - Mar 2025

Vice Coordinator of Kelompok Pengembangan Baturetno

- Assisted the coordinator in planning, organizing, and supervising community development programs for the Baturetno Development Group.
- Coordinated team members by assigning responsibilities, monitoring progress, and ensuring work programs were implemented according to schedule.
- Facilitated communication and collaboration with local stakeholders, schools, and community organizations to support program implementation.
- Evaluated program outcomes and coordinated follow-up activities to improve the effectiveness and sustainability of community initiatives.

BIMITS TALKS 2024

Oct 2024

Head of Secretarial Division

- Led the Secretarial Division by coordinating administrative activities before, during, and after the event.
- Prepared and managed official documents, correspondence, meeting minutes, and event administration.
- Maintained participant records and ensured the accuracy and completeness of event documentation.
- Coordinated with other divisions to support the smooth planning and execution of the event.

GERIGI ITS 2024

Jul 2024 - Aug 2024

Staff of Mentor

- Assisted new students in adapting to university life and the academic environment.
- Provided guidance on academic activities, campus organizations, and student services.
- Served as a reliable point of contact by addressing questions and providing ongoing support throughout the orientation program.
- Collaborated with fellow mentors and the organizing committee to ensure the orientation activities ran smoothly.

BIMITS IMPROVE SKILLS 2024

May 2024

Head of Secretarial Division

- Led the Secretarial Division by overseeing administrative operations before, during, and after the event.
- Prepared and managed official documents, correspondence, meeting minutes, and event administration.
- Maintained participant records and ensured all administrative documents were accurate and well organized.
- Coordinated with committee members to ensure administrative processes were completed efficiently and on schedule.

BIDIKMISI ITS

Mar 2024 - Jan 2025

Expert Staff of Student Development Division

- Assisted in planning and implementing student development programs for Bidikmisi ITS scholarship recipients.
- Coordinated program activities and managed administrative processes to ensure smooth event execution.
- Prepared meeting materials, documentation, and reports to support program planning and evaluation.
- Facilitated communication and collaboration among committee members and scholarship recipients throughout program implementation.

HIMATIKA ITS

Feb 2024 - Jan 2025

Staff of Internal Affairs

- Organized the Kastrat Discussion program by coordinating a strategic academic discussion on "Optimizing Pre-Exam Learning: Is Tutoring an Effective Preparation Method?"
- Managed the HIMATIKA Birthday Calendar (HBC) by maintaining member birthday databases, creating digital greeting content, and scheduling birthday announcements on HIMATIKA ITS's official Instagram to enhance organizational engagement and appreciation.

Ikatan Mahasiswa Berprestasi Wonogiri

Jan 2024 - Mar 2024

Secretary of Kelompok Pengembangan Baturetno

- Prepared meeting minutes, official documents, and administrative records to support the planning and implementation of community development programs.
- Organized schedules, maintained program documentation, and managed activity records to ensure well-structured and accurate administration.
- Coordinated communication among team members and stakeholders to facilitate the smooth execution of community development initiatives.

AKSI NIRWASITA 4.0

Sep 2023

Staff of Data Center

- Supported the Data Center team during the faculty orientation program for new students.
- Recorded and monitored participant activity points throughout the event.
- Collected, organized, and maintained student data to ensure accurate and up-to-date records.
- Verified data accuracy and managed event documentation to support efficient coordination and program execution.

GERIGI ITS 2023

Aug 2023

Staff of Secretarial

- Prepared and managed official documents, correspondence, and administrative records throughout the orientation program.
- Recorded meeting minutes and maintained documentation to ensure accurate tracking of committee decisions and activities.
- Organized participant data and verified administrative information to support efficient event operations.
- Coordinated with committee members to facilitate document distribution and ensure timely completion of administrative tasks.

SCETCH 2023

May 2023 - Sep 2023

Staff of Event Mobile Legend

- Developed the event rundown, Terms of Reference (ToR), technical guidelines, and Memorandum of Understanding (MoU) for the Mobile Legends tournament.
- Prepared technical meeting materials, competition regulations, and operational documents to ensure clear communication with participating teams.
- Coordinated with the event committee to align tournament schedules, match procedures, and technical requirements before and during the competition.
- Monitored the implementation of the tournament flow and assisted in resolving operational issues to ensure the competition was conducted according to the planned schedule.

BIMITS

Mar 2023 - Mar 2024

Dewan Perwakilan Departemen (DPD)

- Disseminated important information regarding Bidikmisi and KIP-K scholarships to students in the Mathematics Department, ensuring timely and accurate communication.
- Served as a liaison between the BIMITS executive board and scholarship recipients by conveying announcements, collecting feedback, and facilitating effective communication.

ACHIEVEMENTS

- **Penghargaan Mahasiswa Berprestasi Kabupaten Wonogiri 2025** 🏆 (2025): Selected as a recipient of the 2025 Wonogiri Outstanding Student Award by the Wonogiri Regency Government in recognition of academic excellence, extracurricular achievements, and community contributions. Contributed to community development through IMAPRES Goes to School, rice husk briquette training for local women's groups (PKK), stock investment literacy programs for youth organizations, and other sustainable community empowerment initiatives.
- **Penghargaan Mahasiswa Berprestasi Kabupaten Wonogiri 2024** 🏆 (2024): Selected as a recipient of the 2024 Wonogiri Outstanding Student Award by the Wonogiri Regency Government for academic excellence, extracurricular achievements, and community contributions. Served as Vice Coordinator of the Baturetno Development Group, leading the planning and implementation of community development programs, including Youth Literacy Leadership x Sehat Berani Pintar, IMAPRES Goes to School, parenting workshops for PKK members, and educational outreach initiatives to promote literacy, leadership, and youth development.
- **Penghargaan Mahasiswa Berprestasi Kabupaten Wonogiri 2023** 🏆 (2023): Selected as a recipient of the 2023 Wonogiri Outstanding Student Award by the Wonogiri Regency Government for academic excellence, extracurricular achievements, and community contributions. Served as Secretary of the Baturetno Development Group, managing administrative documentation and supporting community development initiatives, including IMAPRES Goes to School, environmental conservation through the 1,000 Trees Planting

Program, eco-enzyme training for PKK members, elementary school mentoring, and extreme poverty data collection in collaboration with the local government.